



SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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HOSPITAL PATIENT VITAL SIGNS MONITOR DASHBOARD

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CSA0704 – COMPUTER NETWORKS

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INTRODUCTION



- A **digital dashboard** used to monitor a patient's vital signs.
- Displays important health parameters in **real time**.
- Monitors **heart rate, blood pressure, temperature, SpO₂, and respiratory rate**.
- Helps doctors and nurses **track patient health easily**.
- Provides **quick alerts** when readings become abnormal.
- Improves **continuous patient monitoring and timely medical care**.

ABSTRACT

The Hospital Patient Vital Signs Monitor Dashboard is a digital system designed to monitor and display patients' vital signs in real time. It tracks parameters such as heart rate, blood pressure, temperature, SpO₂, and respiratory rate. The dashboard provides clear visual information and alerts for abnormal readings, helping healthcare professionals monitor patients efficiently and provide timely care.

PROBLEM STATEMENT

- Manual monitoring of patient vital signs can be **time-consuming**.
- Continuous changes in vital signs may not be noticed immediately.
- Patient health data can be **difficult to organize and track**.
- Healthcare staff need a **centralized monitoring system**.
- A real-time dashboard is needed to **display vital signs and identify abnormal readings quickly**.
- The system aims to improve **efficient patient monitoring and timely response**.

OBJECTIVES

- To develop a **real-time patient vital signs monitoring dashboard**.
- To monitor **heart rate, blood pressure, temperature, SpO₂, and respiratory rate**.
- To display patient health data in a **simple and user-friendly interface**.
- To identify **abnormal vital sign readings** quickly.
- To provide **alerts for critical changes** in patient conditions.
- To support healthcare professionals in **efficient patient monitoring**.

SYSTEM ARCHITECTURE

- **Patient → Sensors → Data Collection → Processing → Database → Dashboard → Doctor/Nurse**
- **Patient:** Provides the vital sign data.
- **Sensors:** Measure heart rate, blood pressure, temperature, SpO₂, etc.
- **Data Collection:** Receives readings from the sensors.
- **Processing:** Checks and analyzes the collected data.
- **Database:** Stores patient information and vital sign history.
- **Dashboard:** Displays real-time readings using numbers, graphs, and status indicators.
- **Doctor/Nurse:** Monitors the dashboard and responds to abnormal readings.

DASHBOARD FEATURES

- **Patient Information** – Displays patient ID, name, age, and room number.
- **Real-Time Monitoring** – Shows current vital sign readings.
- **Vital Signs Display** – Heart rate, blood pressure, temperature, SpO₂, and respiratory rate.
- **Status Indicators** – Shows readings as normal or abnormal.
- **Alerts & Notifications** – Gives warnings when readings are outside the set range.
- **Graphs & Trends** – Displays changes in vital signs over time.
- **Patient History** – Stores and displays previous readings.
- **User-Friendly Interface** – Easy for doctors and nurses to monitor patients.

IMPLEMENTATION

- **Data Collection:** Collect patient vital signs using sensors or manually entered data.
- **Data Processing:** Process and validate the collected readings.
- **Database:** Store patient details and vital-sign history securely.
- **Dashboard Development:** Display readings using charts, graphs, and status indicators.
- **Alert System:** Generate alerts when readings cross predefined limits.
- **Real-Time Update:** Continuously update the dashboard with the latest readings.
- **User Access:** Allow authorized healthcare staff to view and monitor patient data.

MODULE-1 HARDWARE AND DATA COLLECTION

- **Sensors:** Measure heart rate, blood pressure, temperature, SpO₂, and respiratory rate.
- **Microcontroller:** Collects and processes sensor readings.
- **Data Acquisition:** Receives vital-sign data from the sensors.
- **Communication:** Transfers collected data to the monitoring dashboard.
- **Patient Identification:** Links the readings with the correct patient.
- **Data Storage:** Stores the collected readings for further monitoring and analysis.



MODULE – 2 DASHBOARD AND SOFTWARE



- **Dashboard Interface:** Displays patient vital signs clearly.
- **Real-Time Monitoring:** Shows the latest readings continuously.
- **Data Visualization:** Uses graphs and charts to show vital-sign trends.
- **Alert System:** Indicates abnormal or critical readings.
- **Patient Management:** Displays patient details and monitoring history.
- **Database:** Stores patient information and vital-sign records.
- **Software Processing:** Analyzes and updates the collected data.

MODULE-1- APPLICATION

```
1 # Module 1 - Hardware and Data Collection
2
3 print("=== PATIENT VITAL SIGNS DATA COLLECTION ===")
4
5 name = input("Enter Patient Name: ")
6 age = int(input("Enter Age: "))
7
8 heart_rate = int(input("Enter Heart Rate (BPM): "))
9 blood_pressure = input("Enter Blood Pressure (mmHg): ")
0 temperature = float(input("Enter Temperature (°C): "))
1 spo2 = int(input("Enter SpO2 (%): "))
2 respiratory_rate = int(input("Enter Respiratory Rate (/min): "))
3
4 print("\n=== COLLECTED PATIENT DATA ===")
5 print("Patient Name      :", name)
6 print("Age                :", age)
7 print("Heart Rate          :", heart_rate, "BPM")
8 print("Blood Pressure       :", blood_pressure, "mmHg")
9 print("Temperature          :", temperature, "°C")
0 print("SpO2                 :", spo2, "%")
1 print("Respiratory Rate     :", respiratory_rate, "/min")
2
3 print("\nData collection completed successfully!")
```

MODULE -1 -APPLICATION

```
=== PATIENT VITAL SIGNS DATA COLLECTION ===  
Enter Patient Name: JAYA  
Enter Age: 18  
Enter Heart Rate (BPM): 120  
Enter Blood Pressure (mmHg): 36  
Enter Temperature (°C): 36  
Enter SpO2 (%): 32  
Enter Respiratory Rate (/min): 120  
  
=== COLLECTED PATIENT DATA ===  
Patient Name      : JAYA  
Age               : 18  
Heart Rate        : 120 BPM  
Blood Pressure    : 36 mmHg  
Temperature       : 36.0 °C  
SpO2              : 32 %  
Respiratory Rate  : 120 /min  
  
Data collection completed successfully!  
  
=== Code Execution Successful ===|
```

MODULE-2-APPLICATION

```
1 #include <stdio.h>
2
3 int main() {
4     char name[50];
5     int age, heartRate, spo2, respiratoryRate;
6     float temperature, systolic, diastolic;
7
8     printf("Enter Patient Name: ");
9     scanf("%s", name);
10
11     printf("Enter Age: ");
12     scanf("%d", &age);
13
14     printf("Enter Heart Rate (BPM): ");
15     scanf("%d", &heartRate);
16
17     printf("Enter Blood Pressure (Systolic/Diastolic): ");
18     scanf("%f/%f", &systolic, &diastolic);
19
20     printf("Enter Temperature (C): ");
21     scanf("%f", &temperature);
22
23     printf("Enter SpO2 (%): ");
24     scanf("%d", &spo2);
25
26     printf("Enter Respiratory Rate: ");
27     scanf("%d", &respiratoryRate);
```

```
printf("\n==== PATIENT VITAL SIGNS DASHBOARD ==== \n");
printf("Patient Name      : %s\n", name);
printf("Age                : %d\n", age);
printf("Heart Rate           : %d BPM\n", heartRate);
printf("Blood Pressure        : %.0f/%.0f mmHg\n", systolic, diastolic);
printf("Temperature           : %.1f C\n", temperature);
printf("SpO2                  : %d%%\n", spo2);
printf("Respiratory Rate       : %d/min\n", respiratoryRate);

printf("\nStatus:\n");

if (heartRate < 60 || heartRate > 100)
    printf("Heart Rate           : ABNORMAL\n");
else
    printf("Heart Rate           : NORMAL\n");

if (spo2 < 95)
    printf("SpO2                  : ABNORMAL\n");
else
    printf("SpO2                  : NORMAL\n");

if (temperature < 36 || temperature > 37.5)
    printf("Temperature           : ABNORMAL\n");
else
    printf("Temperature           : NORMAL\n");

if (respiratoryRate < 12 || respiratoryRate > 20)
    printf("Respiratory Rate       : ABNORMAL\n");
else
    printf("Respiratory Rate       : NORMAL\n");

return 0;
}
```

MODULE-2-APPLICATION

```
Enter Patient Name: MADHU
Enter Age:
18
Enter Heart Rate (BPM): 120
Enter Blood Pressure (Systolic/Diastolic):
180
Enter Temperature (C): 36
Enter SpO2 (%): 32.2
Enter Respiratory Rate:
===== PATIENT VITAL SIGNS DASHBOARD =====
Patient Name      : MADHU
Age               : 18
Heart Rate        : 120 BPM
Blood Pressure    : 180/0 mmHg
Temperature       : 36.0 C
SpO2              : 32%
Respiratory Rate  : 0/min

Status:
Heart Rate        : ABNORMAL
SpO2              : ABNORMAL
Temperature       : NORMAL
Respiratory Rate  : ABNORMAL

=== Code Execution Successful ===
```



Engineer to Excel



WORKSPACE

Overview

Patients

Alerts

WEB PAGE



CLINICAL DASHBOARD

Vital Signs Overview

Real-time monitoring across all admitted patients. Sorted by severity.

Search patients...

+ New Patient

TOTAL PATIENTS

3



STABLE

0

3 awaiting data



WARNINGS

0



CRITICAL

0



Patient Roster

3 SHOWN

Alerts

LIVE

JU

Jane UTest

40 yrs · 202

• NORMAL

No readings recorded yet

AWAITING DATA



L

Lakshana

18 yrs · Female · ICU-5

• NORMAL

No readings recorded yet

AWAITING DATA



CB

Critical Bob

60 yrs

• NORMAL

No readings recorded yet

AWAITING DATA



All patients stable

No abnormal readings detected.

RESULT AND DISCUSSION

- The dashboard successfully **displays patient vital signs in real time.**
- Heart rate, blood pressure, temperature, SpO₂, and respiratory rate can be **monitored easily.**
- **Graphs and status indicators** make changes in vital signs easy to understand.
- The **alert system** helps identify abnormal readings quickly.
- Patient data can be **stored and reviewed** for monitoring trends.
- Overall, the system provides **efficient and user-friendly patient monitoring.**

ADVANTAGES AND APPLICATION

Advantages

- Real-time patient monitoring
- Quick detection of abnormal readings
- Easy-to-use dashboard
- Reduces manual monitoring
- Supports timely medical decisions

Applications

- Hospitals and wards
- ICU and critical care
- Emergency departments
- Remote patient monitoring
- Healthcare research

CHALLENGES AND LIMITATIONS

- **Sensor Accuracy:** Sensors may sometimes provide inaccurate readings.
- **Real-Time Connectivity:** Network failure can interrupt data updates.
- **System Reliability:** Hardware or software failures may affect monitoring.
- **Data Security:** Patient information needs strong privacy protection.
- **Maintenance:** Sensors and devices require regular maintenance.
- **Limited Functionality:** The dashboard monitors vital signs but does not diagnose diseases.

FUTURE SCOPE

- **IoT Integration** – Connect medical sensors for continuous real-time monitoring.
- **AI-Based Analysis** – Use AI to identify unusual vital-sign patterns.
- **Cloud Storage** – Store patient data securely for easy access.
- **Mobile Application** – Allow healthcare staff to monitor patients remotely.
- **Advanced Alerts** – Provide early warnings for critical changes.
- **Predictive Monitoring** – Analyze trends to support early intervention.

CONCLUSION

- The **Hospital Patient Vital Signs Monitor Dashboard** provides an efficient way to monitor patient health.
- It displays **vital signs in real time** through a simple interface.
- Alerts help identify **abnormal readings quickly**.
- Graphs and stored data support **easy tracking of patient conditions**.
- The system can improve **monitoring efficiency and timely medical response**.
- Future technologies such as **IoT, AI, and cloud computing** can further enhance the system.

REFERENCES

- **World Health Organization (WHO) – Digital Health and Health Technologies**
- **Centers for Disease Control and Prevention (CDC) – Vital Signs**
- **Arduino Documentation – Sensors and IoT Projects**
- **National Institute of Biomedical Imaging and Bioengineering (NIBIB) – Medical Devices and Monitoring**
- **Research articles on Patient Vital Signs Monitoring Systems and Healthcare Dashboards**



THANK YOU!